Ayush Yumlembam

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PROFESSIONAL SUMMARY

- ME-Design from BITS, Pilani. ME overall CGPA of 9.53/10.0.
- 10.6 years' experience in gas turbine and aero-engine structural analysis at GE, Safran, and GKN — including projects for **GE Oil & Gas (PGT-25/LM-2500 Low Pressure Gas Turbine static parts), Safran-Snecma (CFM LEAP-1A Airbus A320neo 3D woven composite fan blade, part of the Safran/GE joint program), and GKN (Embraer E190-E2 Turbine Exhaust Casing – TEC, part of Pratt & Whitney PW1900G engine)**.
- Relocation Preference: Open to Asia-Pacific, Europe or Australia.
- Past roles were **Senior Engineer** and **Acting Global Analysis Lead** at GKN Hot Structures Aero-engines team. Reported directly to Engineer-in-Charge (EiC), who is the delegated approver for **EASA**.
- **Seeking re-entry:** Planned career break (Jan 2023 - present) due to a chronic lumbar disc condition, fully resolved as of early 2026 - during which I independently pursued advanced FEA theory, open-source solver frameworks, and fracture mechanics - actively seeking full-time re-entry into structural CAE/FEA engineering roles.
- **Targeting advanced CAE/Structural Analysis roles** with a focus on certification workflows, durability assessment, and computational mechanics. Open to long-term research-driven engineering positions.
- **Current Technical Interests:** Finite Element Method (FEM) solver development, elastic/plastic fracture mechanics, additive manufacturing process modeling, composite material behavior, and open-source CAE toolchains (FEniCSX, Code-Aster).
- Operated a small workshop-based design, fabrication, and welding setup (setup is closed).
- Self-taught in stick welding and fabrication; bridges theoretical CAE with practical manufacturing constraints, weld defect analysis, and residual stress evaluation.
- Responsibilities at GKN included performing **PW1000G (Bombardier/Embraer/Mitsubishi)** Turbine Exhaust Casing (TEC) structural strength and durability analyses for **FAA certification**, automation scripts updates, report reviews, continuous improvement, production support, design improvements, non-conformance support, service manual limits assessment, understanding FAA requirements for refinement of existing assessment methods, methods improvement, tracking first-time-right and on-time delivery. As Global Analysis lead, was involved additionally in, interviewing candidates, work allocation, imparting training, project planning, and minor administrative activities for a team of 3 to 6 members. Attending customer-initiated product reviews and meetings related to air-worthiness issues.

- **Good knowledge of FAA requirements**.
- Experienced in static-structural, impact (beginner), thermal, modal, harmonic, creep, LCF life, HCF life and crack propagation analyses. Additive manufacturing (AM) trainings and literature studies done.
- Worked on aero-engine TEC, woven composite fan blade, woven composite OGV and laminated composite fan casing. Also, worked on gas turbine casing, shroud, nozzle, strut, support ring and exhaust plenum.
- Promoted from Team Lead (L2) to Global Analysis Lead (L3) in 2018.
- **Awarded GKN best team of the year in 2017 for successful completion of FAA certification.**
- Secured second prize in 'Siddhi', a national level robotics competition during APOGEE 2006, at BITS, Pilani, Master's Degree. Worked on 'automated storage and retrieval system' and 'robotic gripper' projects in college. Worked on implementation of automated conveyor system for HVAC assembly in industry.
- Familiar with programming languages like Fortran-77, Matlab, C++, C, Linux Shell, AWK, Jupyter-Notebook, and Python.
- Self-trained in FEniCSX, Salome-Meca/Code-Aster, and have foundational exposure to FreeCAD/CalculiX.
- Hands on experience in PWLIFE, NASGRO, ANSYS Classic, APDL, ANSYS Workbench, Hypermesh, ABAQUS, SAMCEF, Patran, LS-DYNA, Catia V4, PRO-E 2000i, Auto Cad 2000, and UG NX-4&6. Trained in NX Simcenter meshing, and Franc-3D.
- Familiar with Geometric Dimensioning and Tolerancing. Good in MS Word, MS Excel, and Power Point.

PERSONAL SUMMARY

- Interests: Welding, carpentry, electronics, hiking, and martial arts.
- Currently based in Bengaluru, India, with family. Originally from Manipur, India.

ENGINEERING AND ENTREPRENEURIAL EXPERIENCE

Entrepreneurial Venture & Career Transition Period:

- **Aug-2020 to Present, Imphal & Bengaluru, India**

  • Self-Employed Founder (Aug 2020 - Dec 2022): Independently operated a small workshop-based (Imphal, Manipur) design and welding setup focused on small-scale fabrication, custom engineering solutions, and technical training. Managed end-to-end planning and tooling setup under pandemic constraints. Operations were paused due to regional economic constraints and a chronic lumbar disc condition that limited prolonged desk/standing work.

  • Planned Career Break & Medical Recovery (Jan 2023 - Present), due to lumbar disc condition: Focused on targeted medical treatment and structured physiotherapy. Condition is now well-managed; fully cleared for full-time engineering duties as of early 2026.

  • Technical Upskilling (2023 - Present): Independently studied open-source FEA frameworks (FEniCSX, Salome-Meca/Code-Aster), non-linear FEA theory, fracture mechanics, and computational solver literature.

  • Maintained technical engagement through scripting (Python, Code-Aster, FEniCSX).

  • Industry Re-Entry Readiness: Actively seeking to return to full-time structural CAE/FEA engineering roles, bringing refreshed theoretical knowledge, hands-on familiarity with open-source solver frameworks, and extensive industry certification experience.

Structural CAE experience in aero-engines and gas turbines in chronological order:

- **27-Apr-2015 to 25-Aug-2020, GKN Aerospace Engine Systems India Pvt. Ltd., Bangalore**
  Sr. Engineer (Aero-engines Team)

  TEC (turbine exhaust casing) weld analysis: This TEC is a **GKN-produced sub-assembly** for the **Pratt & Whitney PW1900G engine**, used on the **Embraer E190-E2 aircraft**. The robot TIG/plasma/laser welded locations in the TEC are possible failure locations. Cracks usually initiate at the surface of non-weld locations. At weld locations, cracks are assumed to be already present (following a damage tolerant approach), and the cracks then propagate through the material leading to failure. For welds (and for some castings with defects), a damage tolerant LEFM (linear elastic fracture mechanics) crack propagation approach is used to find out the service life of the TEC. NASGRO in conjunction with in-house Python scripts are used to find the service life of the welds. Thermal and mechanical LCF loads are dominant and used for the TEC Certification analyses. ANSYS is used for structural stress evaluation. Tools used are Hypermesh, ANSYS, APDL scripts, in-house Python scripts and NASGRO. Trained in Franc-3D.

  TEC LCF and Strength analyses: LCF crack initiation life (using PWLIFE tool) and Strength analyses are also done to evaluate crack initiation life and strength requirements for Certification using in-house tools and ANSYS. The TEC is also assessed for degraded strength (non-linear elasto-plastic assessment), modal behaviour, creep life and HCF life (due to harmonic forcing). Non weld locations checked for crack initiation life using P&W tool PWLIFE.

  Responsibilities: Contribute individually as a team member by generating and releasing reviewed FAA Certification reports. Assist in automation script development by providing inputs and minor code updates. Continuous improvement activities in the areas of method improvement, process improvement, first-time-right and on-time-delivery. Also actively involved in report reviews, production support, design improvements, non-conformance support, service manual limits assessment, and understanding FAA requirements for refinement of existing assessment methods. As **Acting Global Analysis lead (temporary position)**, was involved in, interviewing candidates, work allocation, imparting training, project planning, and minor administrative activities for a team of 3 to 6 members. Attending customer-initiated product reviews and meetings related to airworthiness issues arising out of escapes.

- **7-Apr-2011 to 17-Apr-2015, SAFRAN ENGINEERING SERVICES INDIA PVT. LTD., Bangalore**
  Sr. Mechanical Engineer (CAE, Aero-engine Fan Blade Development Team)

  Aero-engine fan blade non-linear static analysis: Project under **Safran-Snecma** for **LEAP-1A Airbus A320neo 3D woven composite fan blade** development — part of the **CFM International GE–Safran joint engine program**. Non-linear static structural analysis done using Patran/Workbench and SAMCEF on woven composite fan blades with disk to find out displacements, stresses and strains. Modal frequencies and shapes were also found out, and Campbell diagram developed to find out whether unwanted excitations will be induced by the engine rotation from start-up to steady state flight conditions. Also worked on LS-DYNA bird impact and fan blade-out pre and post processing. Tools used are Hypermesh, Workbench, Patran, SAMCEF, LS-DYNA.

Outlet guide vanes (OGV) linear structural and buckling analysis: Study done to replace metallic OGV wheel by woven composites. The global model is composed of the Aero-engine fan frame and OGV wheel assembly. Linear static structural analysis done on global and sub-model with Fan Blade Out (FBO) high impact loads to find out displacements, stresses and strains. Modal frequencies and shapes were also found out. In addition, OGV critical buckling loads and mode shapes were found out to predict buckling failure. Bolts were not modelled in the global and sub models.

Spinner and front shroud bolted joint non-linear clearance analysis: Study done to find out the effect of clearance on the centre of gravity of spinner and front shroud when bolts were tightened in sequence. The "centre of gravity changes" due to tightening was the output required from this study.

Major Achievements in the company

- For the period 2013-2014, received A grade during appraisal.
- Successfully did OGV aerofoil hexa meshing in Workbench. This kind of work had no precedent in the company due to difficulty. Received high feedback from client (8.2/10).

- **26-Apr-2010 to 06-Aug-2010, GE INDIA TECHNOLOGY CENTRE PVT. LTD., JFWTC Bangalore Engineer (O&G BEC Gas Turbine Team)**

The project covered **GE Oil & Gas low-pressure turbine casing and diffuser components** for **PGT-25 (LM-2500 derivative) medium-duty gas turbines**.

Gas turbine casing thermal and creep analysis: For turbine casing, thermal analysis and static structural analysis for stress and deformation for validating a modified reduced cooling flow. Thermal analysis done for baseline and scaled thermals, which are data matched to real test thermals. LCF and creep studies were done at the highest temperature and stress locations. Creep strain and relaxation stresses were the important parameters in this project.

Gas turbine new exhaust plenum and diffuser validation: Static and transient thermal and structural analyses for validating a new radial exhaust flow plenum and diffuser assembly (older: axial flow). Modal, centre-line shift, clearance, shipping, lifting, disassembly and vortex shedding excitation analyses were also conducted. LCF and creep studies were done at critical locations. The effect of bleeds on the plenum-diffuser stress and deformation was also found out at steady-state and transient conditions using thermal and CFD results.

- **09-Jul-2007 to 20-Jun-2008, GE INDIA TECHNOLOGY CENTRE PVT. LTD., JFWTC Bangalore Engineer (O&G BEC Machine Design Team)**

Gas turbine exhaust plenum standardization: Static structural and modal analysis of a standardized radial flow plenum which can be retrofitted in different types of turbines of the same class. Critical stresses, mode shapes and frequencies were found out. Modal frequencies were checked against excitation frequencies for resonance.

Gas turbine casing deformation analysis: To find out whether deformation of modified casing will affect the secondary cooling flows and the rotating parts. Critical stresses, clearance and gaps for cooling flows were found out at start-up and steady state. Main challenge was solution non-convergence due to large number of contact elements used.

Gas turbine exhaust casing support ring welding analysis: To find out whether support ring welding will have sufficient strength and low cycle fatigue life. LCF life was found out for discontinuous weld at critical stress points.

Other experiences in chronological order:

- **08-Jan-2007 to 15-Jun-2007, BEHR INDIA LTD., Pune**
  Project Trainee (6 Months ME Practice School Programme, Master's Degree)
  Implementation of conveyor system for Scorpio-W HVAC assembly line. Ergonomic layout and workstation design and troubleshooting of conveyor system for the assembly line.

- **02-Sep-2002 to 01-Sep-2003, LOKTAK HYDROELECTRIC PROJECT (NHPC), Komkeirap, Manipur**
  Graduate Course Apprentice (1 Year Apprenticeship Training)
  Assisting engineers in maintenance of Francis hydro turbines, bearing cooling systems, alternator cooling systems, and electrical transformers.

EDUCATION

- **Aug-2005 to Jun-2007, BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, Pilani**
  Master of Engineering in Design (Mechanical Engineering), 2007
  Overall CGPA: 9.53/10.0.

- **Aug-1997 to Jun-2001, BMS COLLEGE OF ENGINEERING, Bangalore**
  Bachelor of Engineering in Mechanical Engineering, 2001
  Marks Obtained: 65.8 %

- **1995-1997, MANIPUR PUBLIC SCHOOL, Imphal, Manipur State**
  Central Board of Secondary Education 12 Grade Exams, 1997
  Marks Obtained: 66.6 %

- **1994-1995, MANIPUR PUBLIC SCHOOL, Imphal, Manipur State**
  Central Board of Secondary Education 10 Grade Exams, 1995
  Marks Obtained: 60 %

SELECTED ENGINEERING COURSES

- Finite Element Method
- Computer Aided Analysis & Design
- Product Design
- Mechanisms & Robotics
- Dynamics & Vibration
- Mechanical System Design
- Non-Destructive Testing Technology
- Design Projects

SELECTED ENGINEERING COURSE PROJECTS

- 'Implementation of conveyor system for Scorpio-W HVAC assembly line' at Behr India Ltd., Pune.
- 'Design and fabrication of solar cylindrical parabolic collector water heating system' at Kotak Urja Pvt. Ltd., Bangalore.
- 'Design and fabrication of automated storage and retrieval system (ASRS)' at BITS, Pilani.
- 'Design and fabrication of temperature sensing intelligent gripper' at BITS, Pilani.

TECHNOLOGICAL SKILLS

- Self-trained in FEniCSX, Salome-Meca/Code-Aster, and have foundational exposure to FreeCAD/CalculiX.
- FE Analysis solvers: ANSYS, SAMCEF and ABAQUS.
- Special Fatigue tools: PWLIFE (Pratt & Whitney LCF tool) and NASGRO (fatigue crack growth tool).
- FE Processing tools: ANSYS Workbench, ANSYS Classic, Hypermesh, Patran, LS-DYNA and NX Simcenter (for meshing, beginner).
- CAD Modelling tools: UG NX-4&6, PRO-E 2000i, Catia V4 and Auto Cad 2000.
- Application tools: MS Word, MS Excel, and Power Point.
- Programming languages: Fortran-77, Matlab (beginner), C++, C, Linux Shell, APDL, AWK (beginner), Jupyter-Notebook, and Python (beginner).
- Familiar with Geometric Dimensioning and Tolerancing.
- Knowledge of 6-Sigma processes.

ACADEMIC ACHIEVEMENTS

- Secured second prize in 'Siddhi', a national level robotics competition during APOGEE 2006, at BITS, Pilani.
- ME overall CGPA of 9.53/10.0.

REFERENCES

Available upon request. Academic and industry referees (including former GKN Engineer-in-Charge & BITS Pilani Professors) have been pre-contacted and consented to provide formal recommendations.